

Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

Classified Practice

Fundamentals of Random Variables

Q001 Written

For the sum of two dice, construct the probability distribution and find $P(5 \leq X \leq 8)$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Classified Practice

Fundamentals of Random Variables

Q003 **Written**

Construct both source probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Classified Practice

Fundamentals of Random Variables

Q005 **Written**

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Q006 Written**Find EX for all six source random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).**

Classified Practice

Fundamentals of Random Variables

Q007 Written

For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Classified Practice

Fundamentals of Random Variables

Q009 **Written****Solve both source Exercise groups of transformed expectations for coins and for balls.**

Q010 Written

For black balls drawn from the stated box, find EX and VX .

Classified Practice

Fundamentals of Random Variables

Q011 Written

Find the mean and variance for the two source Try random variables (a numbered card and winning tickets).

Q012 Written**Find EX and VX for all five source Exercise random variables.**

Classified Practice

Fundamentals of Random Variables

Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Q014 Written

Find the standard deviation from each source probability table and for white balls drawn from 4 red and 3 white.

Classified Practice

Fundamentals of Random Variables

Q015 Written

Find the standard deviation for both source probability-table exercises.

Q016 Written

For the number of winning tickets in the source draw, find VY and σY for $Y=2X+1$ and $Y=-3X-2$.

Classified Practice

Fundamentals of Random Variables

Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Q018 Written

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

Classified Practice

Fundamentals of Random Variables

Q019 Written

Solve the two source transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Q020 Written**Solve the two source Try problems involving net winnings and determining a,b for a die transformation.**

Classified Practice

Fundamentals of Random Variables

Q021 Written

Solve all four source transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Q022 MCQ**A probability distribution must have probabilities that sum to:****Choose the correct option.****A** 0**B** 1**C** the number of outcomes**D** the mean

Classified Practice

Fundamentals of Random Variables

Q023 MCQ

For two fair dice, $PX = 2$ is:**Choose the correct option.**A $1/6$ B $1/36$ C $2/36$

D 0

Q024 MCQ**The number of heads in three coins can be:****Choose the correct option.****A** 1,2,3**B** 0,1,2,3**C** 0,3 only**D** any integer

Classified Practice

Fundamentals of Random Variables

Q025 MCQ

The expected value is calculated as:**Choose the correct option.**

- A $\sum PX = x$
- B $\sum xPX = x$
- C largest x
- D variance squared

Q026 MCQ**The mean number of tails in four fair coins is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Fundamentals of Random Variables

Q027 MCQ

For a fair die, EX is:**Choose the correct option.**

A 3

B 3.5

C 4

D 6

Q028 MCQ **$E(aX+b)$ equals:****Choose the correct option.****A** aEX **B** $aEX+b$ **C** $EX+b$ only**D** $a+b$

Classified Practice

Fundamentals of Random Variables

Q029 MCQ

In general, EX^2 is:**Choose the correct option.**A $E[X]^2$ B equal to EX

C a weighted square sum

D always zero

Q030 MCQ**If $EX=2$, then $E(3X+2)$ is:****Choose the correct option.****A** 6**B** 8**C** 10**D** 12

Classified Practice

Fundamentals of Random Variables

Q031 MCQ

The variance identity is:**Choose the correct option.**

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D $\sqrt{V}$

Q032 MCQ**Variance can never be:****Choose the correct option.**

A positive

B zero

C negative

D fractional

Classified Practice

Fundamentals of Random Variables

Q033 MCQ

The first step in a variance problem is to:**Choose the correct option.**

- A draw the distribution
- B take a square root
- C ignore probabilities
- D add the values

Q034 MCQ

Standard deviation is:**Choose the correct option.**

- A VX
- B $\sqrt{V(X)}$
- C VX squared
- D EX

Classified Practice

Fundamentals of Random Variables

Q035 MCQ

Standard deviation has units:**Choose the correct option.**

A squared units

B the same units as X

C no units

D units cubed

Q036 MCQ **$V(aX+b)$ equals:****Choose the correct option.****A** $aVX+b$ **B** a squared times VX **C** $|a|VX$ **D** $VX+b$

Classified Practice

Fundamentals of Random Variables

Q037 MCQ

 $\sigma(aX+b)$ equals:**Choose the correct option.**A $a \sigma X$ B $|a| \sigma X$ C $a^2 \sigma X$ D $\sigma X+b$

Q038 MCQ**For $Y=X+3$, the variance is:****Choose the correct option.****A** $VX+3$ **B** VX **C** $3VX$ **D** $9VX$

Classified Practice

Fundamentals of Random Variables

Q039 MCQ

A net prize after a fixed fee is modeled as:**Choose the correct option.**

A $Y = X + b$

B $Y = aX + b$

C $Y = X^2$

D $Y = a + b$ only

Q040 MCQ**If $a > 0$ and $\sigma(aX+b)=3$, then a is:****Choose the correct option.****A** $3\sigma X$ **B** $3/\sigma X$ **C** $\sigma X/3$ **D** $3+b$

Classified Practice

Fundamentals of Random Variables

Q041 MCQ

Changing b in $aX+b$ changes:**Choose the correct option.**

- A variance only
- B standard deviation only
- C the mean only
- D nothing

Q042 MCQ**If $EX=2$ and $EX^2=7$, then VX is:****Choose the correct option.****A** 3**B** 5**C** 7**D** 9

Classified Practice

Fundamentals of Random Variables

Quiz WRITTEN 1 **Concept Quiz · Written**

Find EX and VX for the number of heads in three fair coins.

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

Classified Practice

Fundamentals of Random Variables

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X = 2$ and $EX = 3$, find $\sigma(-5X + 7)$.

Classified Practice

Fundamentals of Random Variables

Quiz MCQ 5 [Concept Quiz](#) · MCQ

If $VX=5$, $V(3X+1)$ is:

Choose the correct option.

A 5

B 8

C 15

D 45

Quiz MCQ 6 Concept Quiz · MCQ**For a fair die EX is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 4

Classified Practice

Fundamentals of Random Variables

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

A probability table total must be:

Choose the correct option.

A 0

B 1

C 2

D the mean

Quiz MCQ 8 Concept Quiz · MCQ

If $\sigma X=4$, $\sigma(X-9)$ is:

Choose the correct option.

A 0

B 4

C 13

D 16

Answer key

Use the question number shown on each page.

- Q001** Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$.
- Q002** Use the sample space or combination to list every possible X and sum the relevant probabilities.
- Q003** Use binomial or hypergeometric counts, then add probabilities for the requested values.
- Q004** For four fair coins $EX=2$; for the hypergeometric draw $EX=4/3$.
- Q005** Use $EX = \sum xPX = x$ after constructing each distribution.
- Q006** Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.
- Q007** Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table.
- Q008** Apply linearity for $aX+b$ and use the weighted square sum for X^2 .
- Q009** For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table.
- Q010** $EX = 6/7$; $VX = 20/49$.
- Q011** Construct each probability table, then use $VX = EX^2 - EX^2$.
- Q012** Use the distribution table for each experiment and the variance identity.
- Q013** The table gives $VX = 24/49$, so $\sigma X = 2 \sqrt{6/7}$.
- Q014** Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table.
- Q015** Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$.
- Q016** Use $V(aX + b) = a^2 VX$ and $\sigma(aX+b) = |a| \sigma X$.
- Q017** Apply the squared multiplier to VX and the absolute multiplier to σX .
- Q018** Use the variance and standard-deviation transformation formulas for each Y .
- Q019** Use $E(aX + b) = aEX + b$ and $\sigma(aX+b) = |a| \sigma X$, with $a > 0$.
- Q020** Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.
- Q021** Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.
- Q022** B
- Q023** B
- Q024** B
- Q025** B
- Q026** B
- Q027** B
- Q028** B
- Q029** C
- Q030** C
- Q031** B
- Q032** C
- Q033** A
- Q034** B
- Q035** B
- Q036** B
- Q037** B
- Q038** B
- Q039** B
- Q040** B
- Q041** C
- Q042** A
- Quiz WRITTEN 1** $EX = 3/2$ and $VX = 3/4$
- Quiz WRITTEN 2** $VY = 32$ and $\sigma Y = 4 \sqrt{2}$
- Quiz WRITTEN 3** 12
- Quiz WRITTEN 4** 10
- Quiz MCQ 5** D
- Quiz MCQ 6** C
- Quiz MCQ 7** B
- Quiz MCQ 8** B

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

Classified Practice

Expected Value and Variance of Multiple Variables

Q001 Written

Given the probability distributions of X and Y in the source tables, find $E(X+Y)$.

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q003 Written

For the two source tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Q004 Written

For the two bags in the source exercise, find the expected total number of red balls drawn.

Classified Practice

Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

Classified Practice

Expected Value and Variance of Multiple Variables

Q007 Written

Two 500-point coins and two 100-point coins are tossed. Find EZ.

Q008 Written**A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .**

Classified Practice

Expected Value and Variance of Multiple Variables

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Q010 Written

In two independent raffles, buy two tickets from a $\frac{2}{10}$ winner raffle and two from a $\frac{1}{6}$ winner raffle. Find the expected prizes.

Classified Practice

Expected Value and Variance of Multiple Variables

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at $1, 2, 3$ and Y uniform on $2, 4, 6$, find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q017 Written

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Q018 Written

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Q020 Written**Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.**

Classified Practice

Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Q022 Written**Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.**

Classified Practice

Expected Value and Variance of Multiple Variables

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q025 Written

If X , Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Q026 MCQ **$E(X+Y)$ equals:****Choose the correct option.****A** $EX-EY$ **B** $EX+EY$ **C** $EXEY$ **D** EX/EY

Classified Practice

Expected Value and Variance of Multiple Variables

Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Choose the correct option.**

- A $EXEYEZ$
- B $EX+EY+EZ$
- C $EX-EY-EZ$
- D 0

Q028 MCQ**The expected heads in one fair coin toss is:****Choose the correct option.****A** 0**B** $1/2$ **C** 1**D** 2

Classified Practice

Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Choose the correct option.**

A 0.5

B 1

C 2

D 4

Q030 MCQ**If $EX=2$ and $EY=5$, $E(X+Y)$ is:****Choose the correct option.****A** 3**B** 7**C** 10**D** 25

Classified Practice

Expected Value and Variance of Multiple Variables

Q031 MCQ

 $E(aX+bY)$ equals:**Choose the correct option.**A $aEX+bEY$ B $abE(XY)$ C $EX+EY$ D $a+b$

Q032 MCQ**The expected score of one fair 100-point coin is:****Choose the correct option.**

- A 25
 - B 50
 - C 100
 - D 200
-
-
-

Classified Practice

Expected Value and Variance of Multiple Variables

Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Choose the correct option.**

A 14

B 21

C 28

D 35

Q034 MCQ**A fixed entry fee changes:****Choose the correct option.**

- A variance only
- B the mean only
- C neither mean nor variance
- D standard deviation only

Classified Practice

Expected Value and Variance of Multiple Variables

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:**Choose the correct option.**A $1/5$ B $2/5$ C $1/2$ D $4/5$

Q036 MCQ **$E(XY) = EXEY$ requires:****Choose the correct option.**

- A $X = Y$
- B independence
- C zero means
- D equal variances

Classified Practice

Expected Value and Variance of Multiple Variables

Q037 MCQ

For three fair coins, EX is:**Choose the correct option.**A $1/2$

B 1

C $3/2$

D 3

Q038 MCQ**For a fair die, EY is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 6

Classified Practice

Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Choose the correct option.**

- A dependent
- B independent
- C identical outcomes
- D impossible

Q040 MCQ**If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:****Choose the correct option.****A** 5**B** 6**C** 1**D** 9

Classified Practice

Expected Value and Variance of Multiple Variables

Q041 MCQ

For independent X,Y, $V(X+Y)$ equals:**Choose the correct option.**A $VX-VY$ B $VX+VY$ C $VXVY$

D 0

Q042 MCQ

The variance of a fair die is:**Choose the correct option.**A $7/2$ B $35/12$ C $35/6$ D $1/4$

Classified Practice

Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Choose the correct option.**A $V(X+Y)$ B $\sqrt{V(X+Y)}$ C $VX+VY$ D $E(X+Y)$

Q044 MCQ**The variance of the sum of two independent fair dice is:****Choose the correct option.**A $35/12$ B $35/6$ C $70/6$ D $49/4$

Classified Practice

Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Choose the correct option.**

- A variance
- B standard deviation
- C neither spread measure
- D both spread measures

Q046 MCQ**For independent X, Y, Z , $E(XYZ)$ equals:****Choose the correct option.****A** $EX+EY+EZ$ **B** $EXEYEZ$ **C** 0**D** $E(XY)+EZ$

Classified Practice

Expected Value and Variance of Multiple Variables

Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Choose the correct option.**A $VXVYVZ$ B $VX+VY+VZ$ C $VX-VY-VZ$

D 0

Q048 MCQ**For a binomial count with $n=5$ and $p=1/6$, EX is:****Choose the correct option.****A** $1/6$ **B** $5/6$ **C** $5/36$ **D** 6

Classified Practice

Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Choose the correct option.**

A 2

B 3

C 4

D 6

Q050 MCQ**For three independent fair dice, $V(\text{sum})$ is:****Choose the correct option.**A $35/12$ B $35/6$ C $35/4$ D $343/8$

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 **Concept Quiz · Written**

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Choose the correct option.

A 7

B 10

C 14

D 24

Quiz MCQ 6 Concept Quiz · MCQ

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Choose the correct option.

- A 7
- B 10
- C 25
- D $1/2$

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

For two independent die rolls, $V(X+Y)$ is:

Choose the correct option.

A $35/12$

B $35/6$

C $35/4$

D $7/2$

Quiz MCQ 8 Concept Quiz · MCQ

For three independent variables, $V(X+Y+Z)$ is:

Choose the correct option.

- A the product of variances
- B the sum of variances
- C the sum of means
- D always zero

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, \text{so } E(X + Y) = 97/6.$ Q016 $VX=2/3, VY=3/4, \text{so } V(X + Y) = 17/12.$ Q025 $E(XYZ) = 23/21=3.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$

and $\sigma = \sqrt{17/12}.$

Q026 B

Q042 B

Q043 B

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$

Q017 $V(X + Y) = 35/12 + 35/12 = 35/6;$

Q027 B

Q044 B

Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$

$\sigma = \sqrt{35/6}.$

Q028 B

Q045 C

Q005 Each mean is $2(0.5) = 1, \text{so}$

Q018 $V = 1/4 + 35/12 = 19/6;$

Q029 B

Q046 B

$E(X + Y + Z) = 3.$

$\sigma = \sqrt{19/6}.$

Q030 B

Q047 B

Q006 $Z = 500X + 100Y, \text{so } EZ = 5001 + 1003/2 = 6500.5$

Q019 $V = 21/4 + 31/4 = 5/4.$

Q031 A

Q048 B

Q007 $EZ = 5001 + 1001 = 600.$

Q020 $V = 5/4 + 35/12 = 25/6;$

Q032 B

Q049 C

Q008 $EZ = 37/2 + 57/2 = 28.$

$\sigma = 5/\sqrt{6}.$

Q033 C

Q050 C

Q009 $EZ = 350 + 25 = 175.$

Q021 $EX=4, VX=8/3; E(XYZ) = 64 \text{ and}$

Q034 B

Quiz WRITTEN 1 -2

Q010 $EZ = 22/10 + 21/6 = 11/15.$

$V(sum) = 8.$

Q035 B

Quiz WRITTEN 2 10

Q011 X and Y are independent, so

Q022 $EX=3, VX=8/3; E(XYZ) = 27 \text{ and}$

Q036 B

Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$

$E(XY) = EXEY = 3/27/2 = 21/4.$

$V(sum) = 8.$

Q037 C

Quiz WRITTEN 4 9/4

Q012 $EX = 8/5, EY = 4, \text{so } E(XY) = 32/5.$

Q023 $E(product) = 7/2^3 = 343/8;$

Q038 C

Quiz MCQ 5 B

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

$V(sum) = 335/12 = 35/4.$

Q039 B

Quiz MCQ 6 B

Q014 $EX = EY = 5/2, \text{so } E(XY) = 25/4.$

Q024 Means are $5/6, 2/3, 1/2;$

Q040 B

Quiz MCQ 7 B

Q015 $E(XY) = EXEY = 2/7.$

$E(product) = 5/18 \text{ and } V(sum) = 5/3.$

Q041 B

Quiz MCQ 8 B